SMART BATTERY MANAGEMENT SYSTEM

Battery Condition Technical Report

REPORT ID: TEST-001
GENERATED: 2026-10-21 08:48:17 UTC

REPORTING PERIOD: 24H
BATTERY BUS: 12.0V Nominal

SPECIFICATION STATUS: UNVERIFIED PROTOTYPE · Profile: **Nominal 12V Lithium-Ion (Unverified Chemistry)** (Unverified Lithium-Ion). Exact chemistry and model are unverified. Provisional safety margins are applied. Do not exceed 45°C ambient or provisional voltage bounds.

SECTION A: OVERALL OBSERVED CONDITION

STATUS: INSUFFICIENT DATA

SPECIFICATIONS UNVERIFIED: Exact cell chemistry and arrangement are unconfirmed for 'Nominal 12V Lithium-Ion (Unverified Chemistry)'. Per ISO 26262 guidelines, universal voltage envelopes cannot be assumed for nominal 12V packs.

SECTION B: LATEST VALID SENSOR MEASUREMENTS

Directly sampled from hardware registers via Firebase Realtime Database (/smart_bms/latest).

Parameter	Measured Value	Hardware / Path	Data Status	Technical Context / Scope
Bus Voltage	**No reading / Stale**	INA219 (I²C 0x40)	UNAVAILABLE	Battery terminal potential across high-side shunt.
Shunt Current	**No reading / Stale**	INA219 (I²C 0x40)	UNAVAILABLE	Positive: Charge / Negative: Discharge across 0.1Ω.
Active Power	**No reading / Stale**	Calculated (V × I)	UNAVAILABLE	Real-time electrical power transferred to load bus.
Ambient Temperature	**No reading / Stale**	DHT11 (1-Wire D7)	UNAVAILABLE	**Enclosure ambient air** near pack (NOT cell core).
Ambient Humidity	**No reading / Stale**	DHT11 (1-Wire D7)	UNAVAILABLE	Relative air humidity inside prototype enclosure.

SECTION C: KEY FINDINGS & OBSERVATIONS

1. Awaiting active sensor stream from NodeMCU ESP8266 controller.

SECTION D: HISTORICAL TRENDS & STATISTICAL EXTREMA

Derived mathematically over current sampled interval(s) for the reporting period.

Historical time series readings are limited for this period. Extrema calculations are omitted to avoid unverified interpolation.

SECTION E: EVIDENCE-BASED RECOMMENDED ACTIONS

#	Action Required	Technical Justification & Verification Target
1	**Obtain verified manufacturer datasheet and configure verified battery profile.**	Adhere to battery manufacturer technical specifications and safety envelopes.
2	**Do NOT guess or assume universal voltage limits for nominal 12V batteries.**	Adhere to battery manufacturer technical specifications and safety envelopes.

SECTION F: DATA QUALITY & SENSING LIMITATIONS

1. **Ambient vs. Core Thermal Measurement:** The DHT11 sensor measures enclosure air temperature adjacent to the pack. It does NOT capture internal electrochemical core cell temperature or localized thermal hotspots.
2. **Unverified Chemistry Constraints:** Exact cell chemistry and manufacturer model have not been laboratory-verified. Consequently, manufacturer-specific C-rate curves, SOC, SOH, and RUL are intentionally withheld to prevent misleading claims.
3. **Single-Node Acquisition:** Telemetry is sampled across I²C (INA219) and 1-Wire (DHT11) on the NodeMCU ESP8266 controller. Stale readings (> 60 seconds) are automatically flagged as unverified.

SECTION G: RETRIEVED TECHNICAL CITATIONS

• Texas Instruments — INA219 Bidirectional Current and Power Monitor Datasheet (SBOS448G)
• Aosong Electronics — DHT11 Humidity & Temperature Digital Sensor Technical Specification
• IEEE Std 1491-2012 — Guide for Selection and Use of Battery Monitoring Equipment in Stationary Applications
• ISO 11898-2 — Road Vehicles — Controller Area Network (CAN) — Part 2: High-Speed Medium Access Unit